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☐ Initial submission ☐ Revised version ☒ Final submission

Life Sciences Reporting Summary

Nature Research wishes to improve the reproducibility of the work that we publish. This form is intended for publication with all accepted life science papers and provides structure for consistency and transparency in reporting. Every life science submission will use this form; some list items might not apply to an individual manuscript, but all fields must be completed for clarity.

For further information on the points included in this form, see Reporting Life Sciences Research. For further information on Nature Research policies, including our data availability policy, see Authors & Referees and the Editorial Policy Checklist.

Experimental design

1. Sample size

Describe how sample size was determined.

In this study we analyzed a total 90 samples with RNAseq. This sample size was sufficient as it was designed to be representative of the common circuits used in Synthetic Biology with triplicate coverage of all samples. The experiment was designed so to have biological triplicates for 4 synthetic constructs transformed in E. coli cells plus control samples corresponding to the test samples. Two strains were selected so that burden could be investigated in two different systems. Finally, two time points were chosen to look at the burden response, 15 and 60 min after induction of gene expression. Thus, DH10B and MG1655 cells transformed with 4 test constructs, their 3 empty plasmid were selected together with empty DH10B as control of untransformed cells.

2. Data exclusions

Describe any data exclusions.

We excluded the RNAseq data for one of the triplicates of the DH10B empty strain as the sample showed lower library abundance and, as a consequence, behaved as an out-lier when compared to the other two replicates.

3. Replication

Describe whether the experimental findings were reliably reproduced.

RNAseq analysis was performed on 3 biological replicates for each sample.
All 90 reaction steps from total RNA to library preparation were successful but the library preparation for the first replicate for DH10 empty cells led to insufficient DNA amount for exhaustive sequencing quality. This sample was then excluded from the analysis. In vivo burden analysis was performed on 2 biological replicates. The biomolecular feedback device characterization and test were performed in 3 biological replicates.

4. Randomization

Describe how samples/organisms/participants were allocated into experimental groups.

Samples were organised into experimental groups based on the aim of the experiment (e.g. for Figure sub-panels). Randomization was not relevant for this study as position of each sample was not likely to influence the outcome.

5. Blinding

Describe whether the investigators were blinded to group allocation during data collection and/or analysis.

No blinding was not use during this study. Blinding was not relevant in the present study as no anticipated result was expected.

Note: all studies involving animals and/or human research participants must disclose whether blinding and randomization were used.

Nature Methods: doi:10.1038/nmeth.4635

6. Statistical parameters

For all figures and tables that use statistical methods, confirm that the following items are present in relevant figure legends (or in the Methods section if additional space is needed).

n/a	Confirmed
☐	☒ The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement (animals, litters, cultures, etc.)
☐	☒ A description of how samples were collected, noting whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒ A statement indicating how many times each experiment was replicated
☒	☐ The statistical test(s) used and whether they are one- or two-sided (note: only common tests should be described solely by name; more complex techniques should be described in the Methods section)
☐	☒ A description of any assumptions or corrections, such as an adjustment for multiple comparisons
☐	☒ The test results (e.g. *P* values) given as exact values whenever possible and with confidence intervals noted
☐	☒ A clear description of statistics including central tendency (e.g. median, mean) and variation (e.g. standard deviation, interquartile range)
☐	☒ Clearly defined error bars

See the web collection on statistics for biologists for further resources and guidance.

Software

Policy information about availability of computer code

7. Software

Describe the software used to analyze the data in this study.	Excel 2010, Python 2.7 and R (DESeq2_1.10.1) were used for data analysis and simulations

For manuscripts utilizing custom algorithms or software that are central to the paper but not yet described in the published literature, software must be made available to editors and reviewers upon request. We strongly encourage code deposition in a community repository (e.g. GitHub). *Nature Methods* guidance for providing algorithms and software for publication provides further information on this topic.

Materials and reagents

Policy information about availability of materials

8. Materials availability

Indicate whether there are restrictions on availability of unique materials or if these materials are only available for distribution by a for-profit company.	no unique materials were used

9. Antibodies

Describe the antibodies used and how they were validated for use in the system under study (i.e. assay and species).	None used

10. Eukaryotic cell lines

a. State the source of each eukaryotic cell line used.	None used
b. Describe the method of cell line authentication used.	None used
c. Report whether the cell lines were tested for mycoplasma contamination.	None used
d. If any of the cell lines used are listed in the database of commonly misidentified cell lines maintained by ICLAC, provide a scientific rationale for their use.	None used

Animals and human research participants

Policy information about studies involving animals; when reporting animal research, follow the ARRIVE guidelines

11. Description of research animals

Provide details on animals and/or animal-derived materials used in the study.	None used

Nature Methods: doi:10.1038/nmeth.4635

Policy information about studies involving human research participants

12. Description of human research participants

Describe the covariate-relevant population characteristics of the human research participants.	None used

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Flow Cytometry Reporting Summary

Form fields will expand as needed. Please do not leave fields blank.

Data presentation

For all flow cytometry data, confirm that:

☒ 1. The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).

☒ 2. The axis scales are clearly visible. Include numbers along axes only for bottom left plot of group (a 'group' is an analysis of identical markers).

☒ 3. All plots are contour plots with outliers or pseudocolor plots.

☒ 4. A numerical value for number of cells or percentage (with statistics) is provided.

Methodological details

5. Describe the sample preparation.	Bacteria were grown in 50 mL of media in 500-mL baffled shake flasks at 37 degrees C for 24 hours. Every hour from 16 to 24 hr, 350 µl of each culture was diluted in 650 µl of PBS and stored on ice in a 96 well microtitre plate. . At the end of the experiment, the PBS-diluted aliquots stored on ice were run on a BD LSR Fortessa X20 flow cytometer to obtain single-cell data.
6. Identify the instrument used for data collection.	Experiments were performed on an BD LSR Fortessa X20 flow cytometer with 96-well plate autosampler attached.
7. Describe the software used to collect and analyze the flow cytometry data.	Data was collected using the BD LSR Fortessa X20 software and analyzed using FlowJo software.
8. Describe the abundance of the relevant cell populations within post-sort fractions.	Cells were gated for shape and size to ensure only bacteria events are captured.
9. Describe the gating strategy used.	Cells were gated during acquisition to ensure that only bacterial cells were acquired based on past experience using the BD LSR Fortessa X20 cytometer.

Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information. ☐

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